

Quiz - 1 Solutions.

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1.

$$y(t) = \int_{t-10}^t \cos(\tau) x(\tau) d\tau.$$

In order to prove that this is LTI, one needs to check for linearity and time invariance.

(a) Linearity: For arbitrary constants a_1 & a_2 , input signals $x_1(t)$ & $x_2(t)$,

$$y_1(t) = \int_{t-10}^t \cos(\tau) x_1(\tau) d\tau$$

$$y_2(t) = \int_{t-10}^t \cos(\tau) x_2(\tau) d\tau.$$

Linear combination yields:

$$y'(t) \triangleq a_1 y_1(t) + a_2 y_2(t)$$

Let i/p to system be $a_1 x_1(t) + a_2 x_2(t)$.

$$\begin{aligned} \Rightarrow y(t) &= \int_{t-10}^t \cos(\tau) (a_1 x_1(\tau) + a_2 x_2(\tau)) d\tau \\ &= a_1 \int_{t-10}^t \cos(\tau) x_1(\tau) d\tau + a_2 \int_{t-10}^t \cos(\tau) x_2(\tau) d\tau \end{aligned}$$

$$= a_1 y_1(t) + a_2 y_2(t) = y'$$

→ Hence system is linear. --- (1)

(b) Time-invariance:

$$y(t-t_0) = \int_{t-t_0-t_0}^{t-t_0} \cos(\tau) x(\tau) d\tau$$

O/p when i/p is shifted by t_0 : i.e.,

$$y'(t) = \int_{t-t_0}^t \cos(\tau) x(\tau-t_0) d\tau$$

$$y(t-t_0) \neq y'(t)$$

Hence the system is not time-invariant

⇒ System is not LTI! --- (1)

(2) $m(t)$: Baseband with bandwidth W .

$$a(t) = m(t) \cos(2\pi f_c t) \quad b(t) = m(t) \sin(2\pi f_c t)$$

$$\langle a, b \rangle = \int_{-\infty}^{\infty} a(t) b(t) dt$$

$$(0.5) \quad \dots = \int_{-\infty}^{\infty} \underbrace{m^2(t) \cos(2\pi f_c t) \sin(2\pi f_c t)}_{x(t)} dt$$

To show that inner product is zero, we

(0.5) -- need to show the baseband of $x(t)$ has no components or that $x(t)$ is passband.

why :

$$X(f) \leftrightarrow x(t) ; x(t) = \int x(t) e^{j2\pi f t} dt$$

If $X(0) = \int x(t) dt = 0$; then it means that

$a(t)$ & $b(t)$ are orthogonal.

hence show $x(t)$ is passband.

$$x(t) = (y(f-f_c) + y(f+f_c)) \cdot 1/2$$

where $m^2(t) = y(t) \leftrightarrow Y(f)$ is baseband with

freq in range $-2W \leq f \leq 2W$ (1)

But $f_c \gg 2W$. $\Rightarrow x(t)$ is passband.

$$3. \quad S(t) = S_I(t) \cos(2\pi f_c t) - S_Q(t) \sin(2\pi f_c t).$$

(a) Upper arm:

Before LPF:

$$x'(t) = 2S_I(t) \cos(2\pi f_c t) \cos(2\pi f_c t + \phi) \\ - 2S_Q(t) \sin(2\pi f_c t) \cos(2\pi f_c t + \phi)$$

Use ~~$\cos(A+B)$~~ =

$$2 \cos\left(\frac{a+b}{2}\right) \cos\left(\frac{a-b}{2}\right) = \cos a + \cos b.$$

$\Rightarrow x'(t) =$

$$2 \sin\left(\frac{a+b}{2}\right) \cos\left(\frac{a-b}{2}\right) = \sin(a) + \sin(b).$$

$$\Rightarrow x'(t) = \cancel{S_I(t)} \cos(4\pi f_c t + \phi) \cos(\phi)$$

$$x'(t) = S_I(t) (\cos(4\pi f_c t + \phi) + \cos(\phi)) \\ - S_Q(t) (\sin(4\pi f_c t + \phi) + \sin(\phi))$$

After LPF:

$$X = S_I(t) \cos \phi - S_Q(t) \sin(\phi) \quad \dots (1)$$

(b) Lower arm: Using same identities:

$$y'(t) = -2S_I(t) \cos(2\pi f_c t) \sin(2\pi f_c t + \phi) \\ - 2S_Q(t) \sin(2\pi f_c t) \sin(2\pi f_c t + \phi)$$

Use

$$\cos(a) - \cos(b) = -2 \sin\left(\frac{a+b}{2}\right) \sin\left(\frac{a-b}{2}\right)$$

$$\sin(a) + \sin(b) = 2 \sin\left(\frac{a+b}{2}\right) \cos\left(\frac{a-b}{2}\right)$$

~~$$y'(t) = 2 S_I(t) \cos 2\pi f_c t$$~~

$$y'(t) = (S_I(t) (\sin(4\pi f_c t + \phi) + \sin(\phi)) + S_Q(t) (\cos(4\pi f_c t + \phi) - \cos \phi))$$

After LPF:

$$Y = S_I(t) \sin \phi + S_Q(t) (-\cos \phi)$$

$$= -S_I(t) \sin \phi + S_Q(t) \cos \phi$$

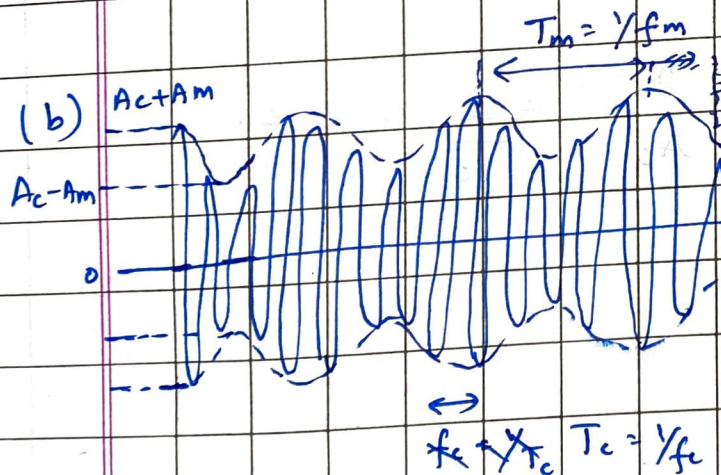
$$- - - (1)$$

(a) $c(t) = A_c \cos(2\pi f_c t)$, $m(t) = A_m \cos(2\pi f_m t)$

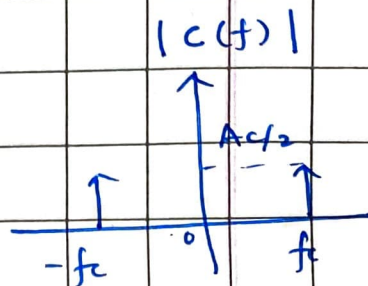
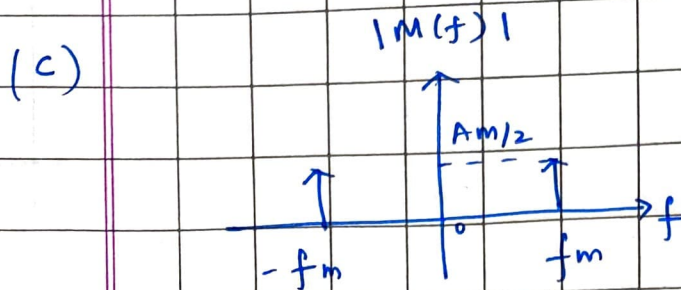
$$s(t) = A_c + A_m \cos 2\pi f_m t \cos 2\pi f_c t$$

(a) $s(t) = A_c \cos 2\pi f_c t + \frac{A_m}{2} \left(\cos(2\pi(f_c - f_m)t) + \cos(2\pi(f_c + f_m)t) \right)$

--- (0.5)

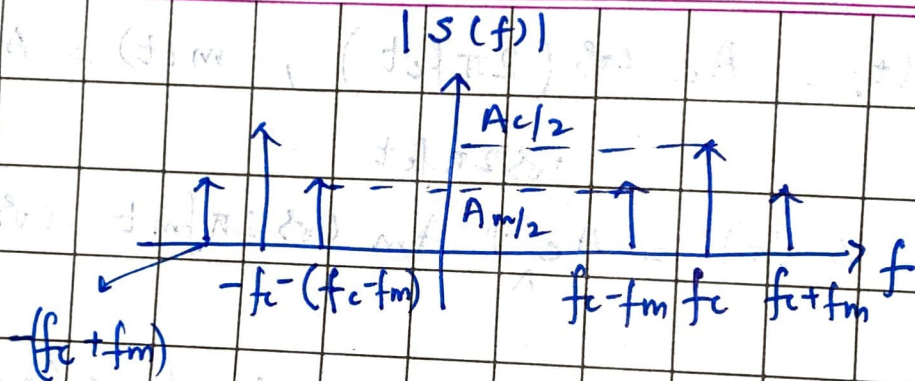


(ensure student has marked everything)



~~--- (0.5)~~

--- (0.5)



----- (1)

(d) Modulation Index:

(ensure everything is marked)

$$\mu = \frac{A_m}{A_c} = \frac{2}{10} = 0.2$$

----- (1)